

Cost Functions, Competitive Behavior, and the Meaning of Profit

Every firm ultimately produces by combining inputs according to some production technology, and its cost function is simply that technology translated into dollar terms: the cheapest way to produce each possible quantity of output, given input prices (Shapiro et al., 2022). Managers rely on cost functions constantly, and not just for bookkeeping. Marginal cost tells a manager whether producing one more unit is worth it at the going price; average cost tells them whether the business is covering itself at current volume; and the shape of the cost curve (whether average cost is still falling because of fixed costs spread over more output, or already rising because of capacity constraints) tells them whether to expand, hold steady, or scale back. Decisions about pricing, output levels, hiring, and even whether to enter or exit a line of business all trace back to a manager's read of the firm's cost structure.

Firms in a perfectly competitive market face a market condition, not a choice: because many firms sell identical products and both buyers and sellers have full information, no single firm can influence the price, so each is a price taker (Shapiro et al., 2022). Given that price, a competitive firm maximizes profit by producing up to the point where marginal cost equals price, since any unit that costs less to produce than it sells for adds to profit, and any unit beyond that point subtracts from it. In the short run a firm may keep producing at a loss as long as price at least covers average variable cost; if not, it is better off shutting down and only losing its fixed costs. In the long run, positive economic profit draws new firms into the market and losses drive firms out, until entry and exit stop and price settles at the minimum of long-run average cost — the point where economic profit is exactly zero (Shapiro et al., 2022).

That zero-profit condition only makes sense once accounting profit and economic profit are separated. Accounting profit is total revenue minus explicit costs, the actual out-of-pocket payments a business makes for wages, rent, materials, and so on. Economic profit goes further, subtracting implicit costs as well: the opportunity cost of resources the owner already has, such as the salary given up to run the business instead of working elsewhere, or the return foregone by tying up one's own capital in the firm rather than investing it elsewhere (Shapiro et al., 2022). Normal profit is the special case where economic profit equals zero — revenue exactly covers both explicit and implicit costs, so the owner is earning exactly what those resources could have earned in their next-best use, no more and no less.

Normal profit is also the one of the three that actually drives how entrepreneurs allocate resources across ventures, because it is defined relative to the opportunity cost of every alternative use of those resources. An entrepreneur deciding whether to keep operating a business, switch industries, or redeploy capital elsewhere isn't asking whether accounting profit is positive; a venture can show a healthy accounting profit while still destroying value if it's below what the owner's time and capital could earn elsewhere. What matters is economic profit relative to the normal-profit benchmark: if economic profit is positive, the venture is beating its next-best alternative and resources should stay or expand there; if economic profit is negative, resources are earning below what they could earn elsewhere and should move; if economic profit is exactly zero (normal profit), the entrepreneur is indifferent between this venture and its next-best alternative. That opportunity-cost logic, not the accounting figure a business reports to the IRS, is what should govern where an entrepreneur puts their time and money.

References

Shapiro, D., MacDonald, D., & Greenlaw, S. A. (2022). Principles of microeconomics (3rd ed.). OpenStax. <https://openstax.org/books/principles-microeconomics-3e/pages/1-introduction>